UTOPIC SENSE

MANUAL

Version 1.1.0

UtopicSense — Version 1.1.0

What’s New

This update focuses on **stability, usability, and persistence**, improving both the workflow inside the Editor and the runtime behavior of sound-driven effects.

🔹 New Features

✔ Refactored SoundEffect System

The SoundEffect component was refactored for better reliability:

- Cleaner separation between audio analysis and visual spawning
- More consistent particle emission timing
- Improved handling of multiple AudioSources

This makes effects more stable across different use cases.

🔹 Improvements

✔ Better Editor Window Stability

- Safer loading and creation of the UtopicSense configuration asset
- Defensive copying of color arrays to avoid accidental data corruption
- Reduced risk of losing settings when reloading scripts

✔ Reduced Dependencies

- The example scene no longer requires **TextMesh Pro**
- Standard Unity UI Text (Legacy) can be used instead
- This reduces external dependencies and keeps integration lightweight

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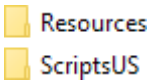
🔹 Notes for Users

- Visual settings are global and shared across scenes
- Scene-specific changes apply only to SoundEffect components
- The example scene UI text is for demonstration purposes only

How to install

After donwloading the package be sure that the **Resources** and **ScriptsUS** folders are inside your **Assets folder package**. Copy the folders mentioned before into your Assets folder in your project if they are not in the correct place.

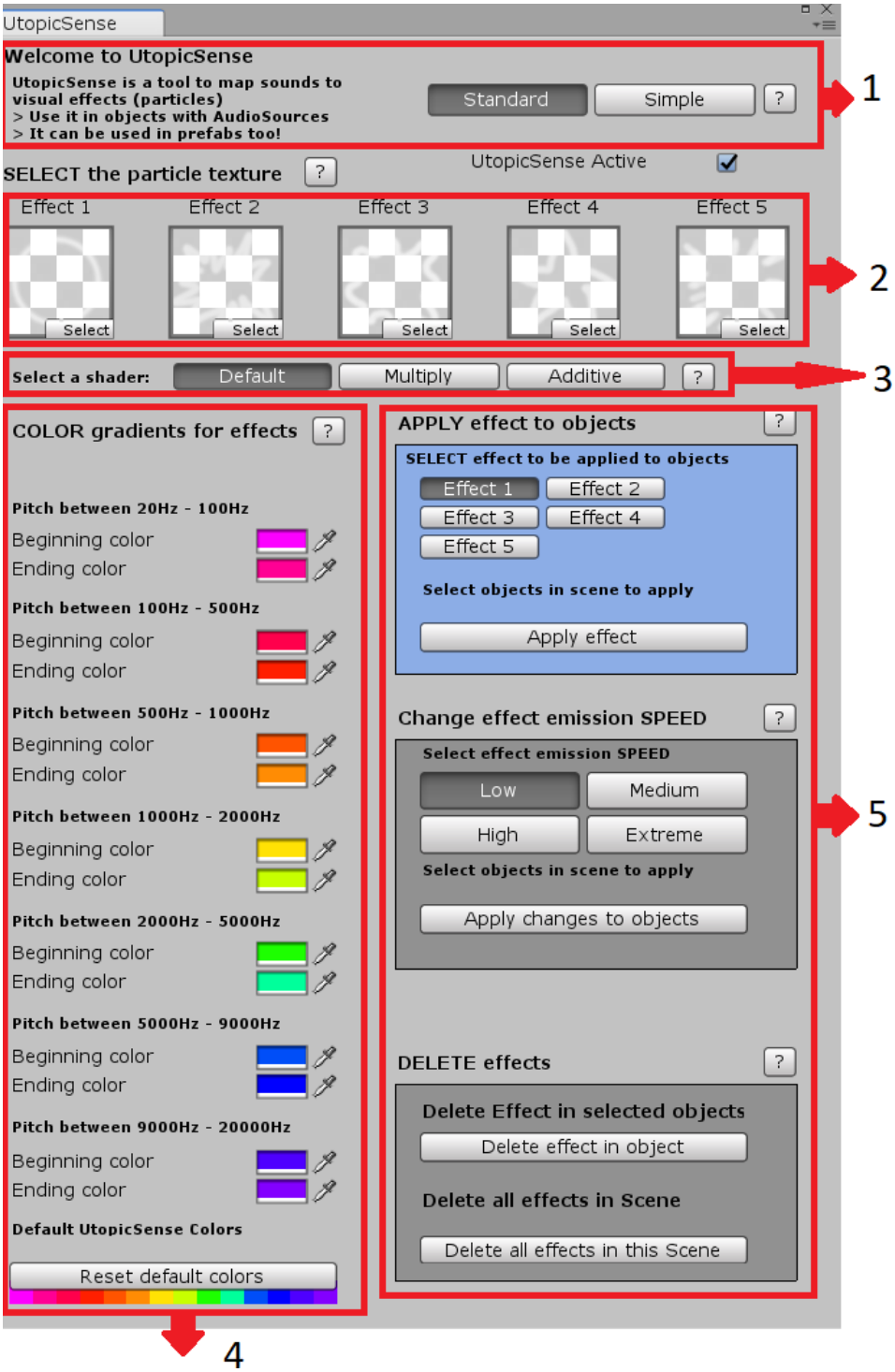


After Unity compiling the scripts you can access the UtopicSense through **Window > UtopicSense**. This is how the interface looks like, you can attach it how you prefer in your Unity interface.

About Utopic Sense

Utopic Sense is a tool for Unity to make it easier for creators to add visual effects to their sound effects in Unity. This device is now focusing on 2D games, but we will further test for 3D games for future versions. The tool comes with "?" buttons to help you navigate and explain each of the functionalities. Here too you can find helpful information about the software.

INTERFACE FEATURES:

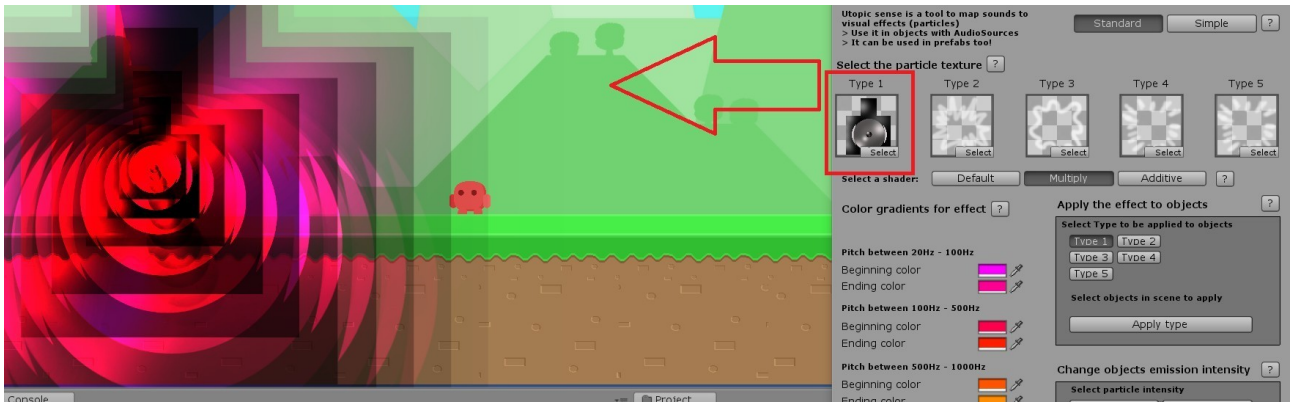


1 - Standard and simple modes

In the upper right side of the program interface, there is the mode selection, which corresponds to UtopicSense being in standard or simplified mode. The application uses the dominant sound frequency in the standard application mode to calculate the color mapping on the image. UtopicSense maps all sounds in a single color gradient in the simplified mode and does not consider the frequency in this process. Beneath this area selection, there is the UtopicSenseActive field, where the user can activate or deactivate the synesthesia effects in the project.

2 - Select particle textures

This region provides means for the user to select and customize their effects according to their artistic vision. UtopicSense starts with five standard textures, but each one can be set to any texture within the project.



3 - Shader selection

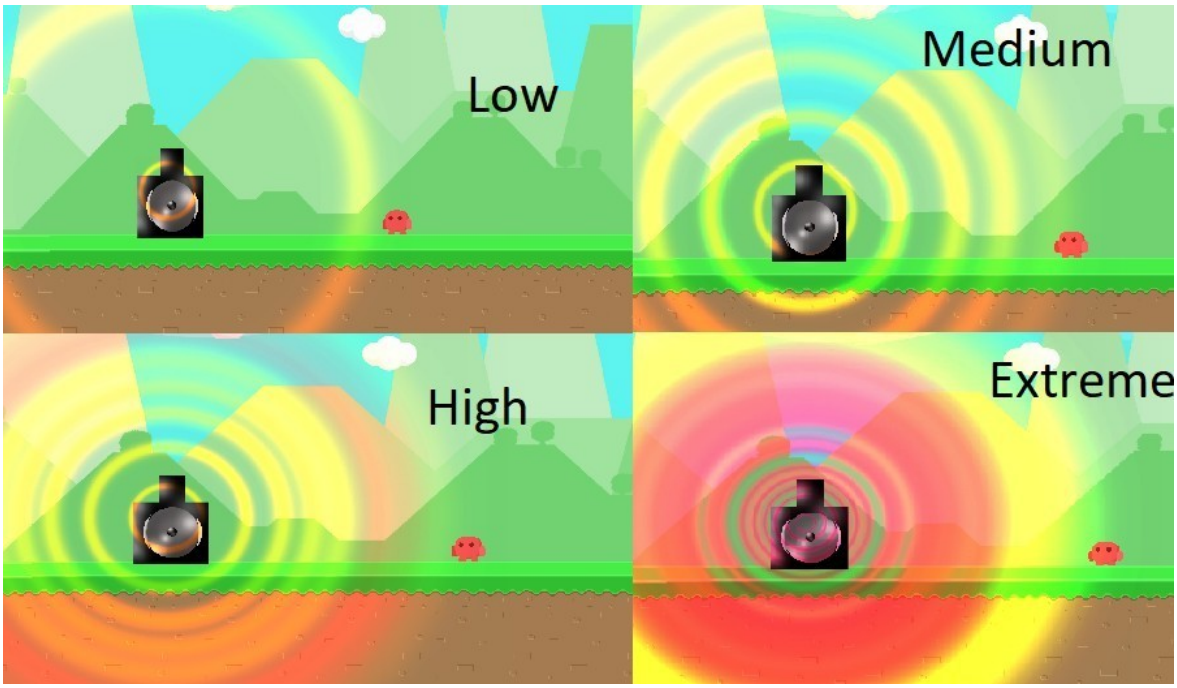
The user can select different shaders to change the visual effect appearance and adapt the effect to different artistic concepts.

4- Color selection

The standard colors are defined by default based on visible light spectrum. The user can modify the colors assigned to frequency bands according to need. After clicking on a color swatch, and RGB color selector, a window will open. Besides the color box, a dropper tool allows selecting any color on the computer screen. If the user wants to return to the default colors of the software, one can use the button "Reset default colors."

5- Apply or remove effect in selected objects

it is up to the game developer to choose and identify which scene bodies will use the visual synesthesia effect. They can attach the particle system by selecting one or more game objects or prefabs and then clicking "Apply effect." The user must select which effect to apply in those objects. Additionally, he can set the effect emission speed and change it in selected items with attached effects.

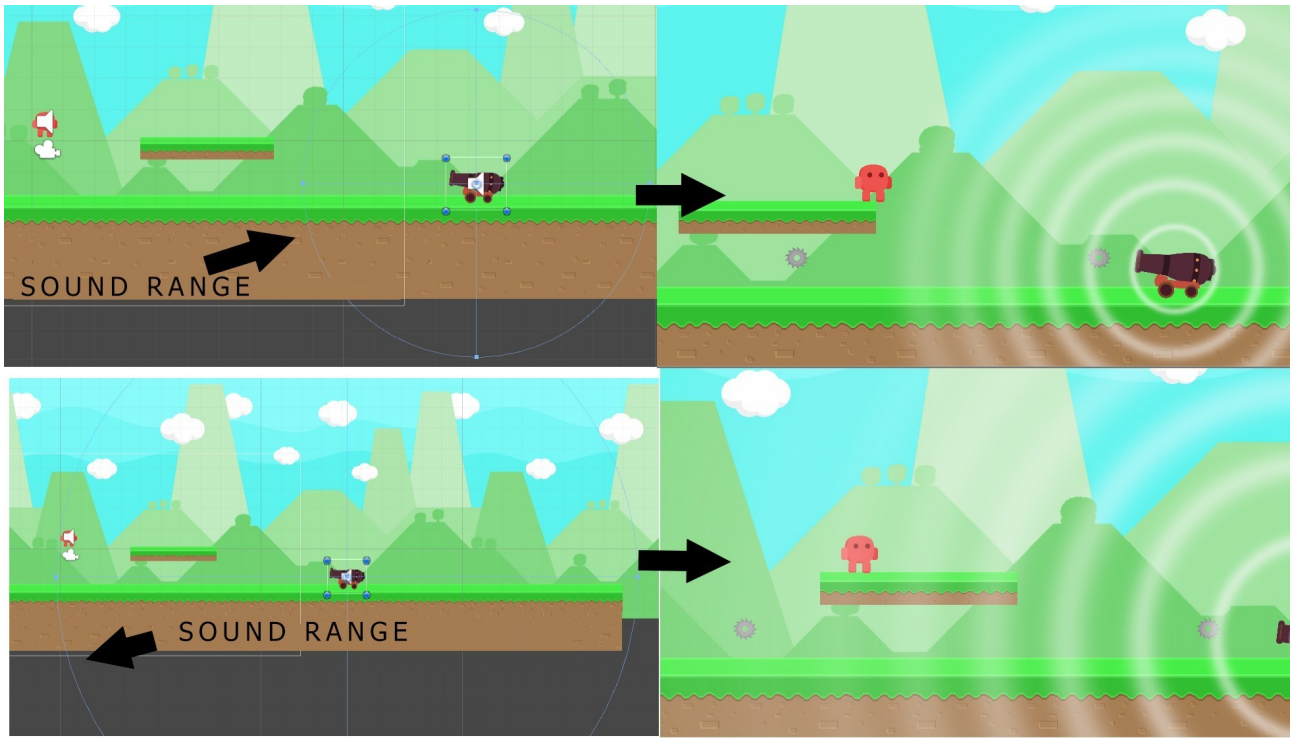


In the same section of the tool's interface, the user can also use the functionality to delete the effect in a single object or delete all effects in the scene.

Other important features

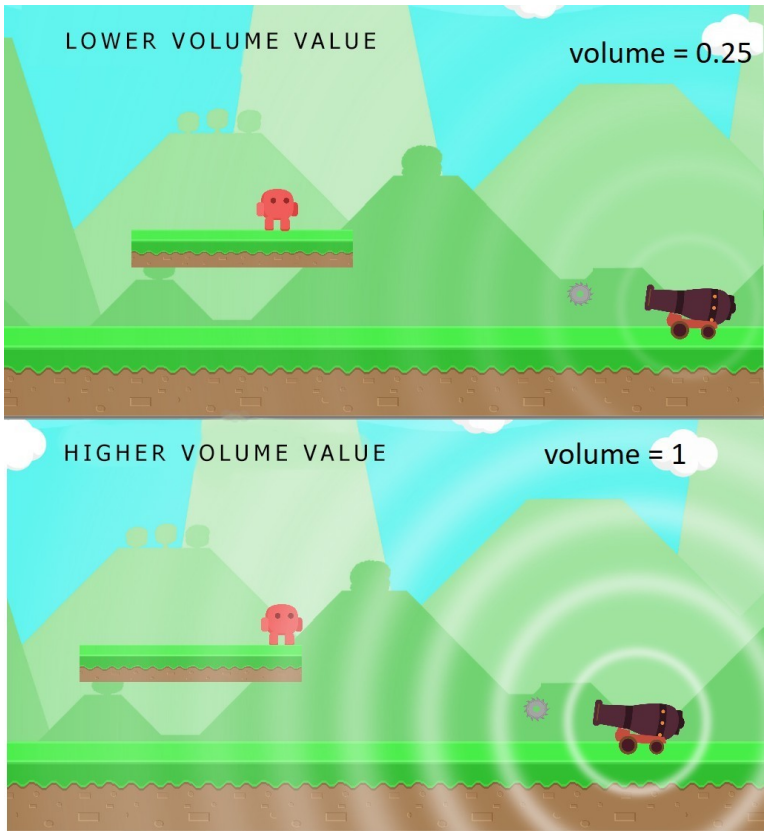
Sound Range

The effect range is proportional to the sound range



Volume

The sound volume maps directly to effect opacity. You can change the volume on the AudioEffect itself on Unity's interface.



Automatic Saving

What you configure in the application will be automatically saved in

`Assets/Resources/PrefabUS/UtopicSense.asset`

If you want to share the configuration with another member of team that also uses UtopicSense you can just share this file.

IEEE Article:

UtopicSense started as an academic project to help developers to map sound effects as particles on scene to favor accessibility. If you are interested in the subject you can read more in the article clicking here.

Source Code:

The scripts of the code and interface are open so you have the freedom to adjust it to your own needs! Hope you have fun with it!

Credits

Software by **Adriel A G Silva**

Special Thanks to

Flávio Coutinho for orienting me on the first version of the project

Gabriel E. O. Gomes for providing the logo design

Hilton Freitas for the emotional support during the start of the development